Partial Fractions

Dhvanit

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Introduction

I have a polynomial $f(x)$ and I want to write $1/f(x)$ as a sum of functions of the form $A/(x+B)$, if possible. Partial fractions, they call it.

$$\frac{1}{x^3-6x^2+11x-6} = \frac{1/2}{x-1} + \frac{-1}{x-2} + \frac{1/2}{x-3} \tag{1}$$

It's not always possible to be able to decompose $1/f(x)$ like that but suppose I have a polynomial $f(x)$ with all roots distinct. Then I can always write:

$$\frac{1}{f(x)} = \sum_{\substack{\alpha\in\mathbb{C} \\ f(\alpha)=0}} \frac{1/f'(\alpha)}{x-\alpha} \tag{2}$$

I won't have to worry about $f'(\alpha) = 0$ since it would imply α is a repeated root. The equality (2) can be justified by multiplying $f(x)$ on both sides (ehh) and the RHS is now a polynomial of degree 1 less than $f(x)$ but it equals the LHS on all the roots of f (... ehh), so it's equal to the LHS always.

But what do I do with $f(x)$ who have repeated roots? If f has a root α which is repeated once, I can't decompose $1/f$ into partial fractions without a term of the form $(Ax+B)/(x-\alpha)^2$. Similarly I'll need to keep terms of the form $p(x)/(x-\alpha)^k$ for more frequently repeated roots.

$$\frac{1}{x^6-10x^5+40x^4-82x^3+91x^2-52x+12} = \frac{(-17x^2+24x-11)/8}{(x-1)^3} + \frac{2x-5}{(x-2)^2} + \frac{1/8}{x-3} \tag{3}$$

> An alternate representation would be to write that term as $\frac{A}{(x-\alpha)} + \frac{B}{(x-\alpha)^2} \cdots$ etc but I choose to not consider it for now because it will end up being a straightforward inter-conversion anyway.

Let $f(x)$ be a polynomial with roots $\alpha_1, \alpha_2, \ldots, \alpha_n$ where the multiplicity of the roots are (respectively) $m_1, m_2, \ldots, m_n$ so that the degree of f is $m_1 + m_2 + \cdots + m_n$. We want to write:

$$\frac{1}{f(x)} = \sum_{i=1}^{n} \frac{p_i(x-\alpha_i)}{(x-\alpha_i)^{m_i}} \tag{4}$$

where $p_i(x)$ is a polynomial of degree $m_i - 1$. The choice of writing $p_i(x-\alpha_i)$ instead of $p_i(x)$ is only for getting convenient expressions later on.

> Using the expansion $p_i(x-\alpha_i) = A_1(x-\alpha_i)^{m_i-1} + A_2(x-\alpha_i)^{m_i-2} + \cdots + A_{m_i-1}(x-\alpha_i) + A_{m_i}$, we can also decompose into simpler partial fractions as described before from $p_i(x-\alpha_i)/(x-\alpha_i)^{m_i}$

The task now is to find $p_i(x)$

Rewrite (4) as:

$$-1 + \sum_{i=1}^{n} p_i(x-\alpha_i)\frac{f(x)}{(x-\alpha)^{m_i}} = 0 \tag{5}$$

Note that $f(x)/(x-\alpha_i)^{m_i}$ is also just a polynomial since α is a root with multiplicity m_i. Let $g_i(x-\alpha_i) = f(x)/(x-\alpha_i)^{m_i}$. Again, defining this way instead of $g_i(x)$ for no other reason than to have cleaner expressions.

$$-1 + \sum_{i=1}^{n} p_i(x-\alpha_i)g_i(x-\alpha_i) = 0 \tag{6}$$

Going on a similar line of thought as for the case of distinct roots, we see that the LHS is a polynomial of degree 1 less than the degree of f. If we can have the roots of LHS be the same as the roots of f, then we know that it must be zero always. But here we need to ensure that, for example, LHS has the root α_i with a multiplicity m_i. In other words, LHS is divisible by $(x-\alpha_i)^{m_i}$ for all $i \in 1, 2, \ldots, n$. But by definition, $g_j(x-\alpha_j)$ is divisible by $(x-\alpha_i)^{m_i}$ for all $j \neq i$. So we need to have the following:

$$(x-\alpha_i)^{m_i} \text{ divides } p_i(x-\alpha_i)g_i(x-\alpha_i) - 1 \qquad \forall i \in \{1, 2, \ldots, n\} \tag{7}$$

or equivalently,

$$x^{m_i} \text{ divides } p_i(x)g_i(x) - 1 \qquad \forall i \in \{1, 2, \ldots, n\} \tag{8}$$

Now this is an independent question for each i, where we are 'given' $g_i(x)$ and we have to 'find' a $p_i(x)$ of degree $m_i - 1$. The coefficients of $g_i(x)$ are 'known'. We have to set the first m_i coefficients of $p_i(x)g_i(x)$ to 0 (or 1) and the m_i coefficients of $p_i(x)$ are our unknowns. This forms a system of linear equations that we know how to solve. But it would be nice to be able to get some final expression too.

$$\sum_{j=0}^{k} p_i[j] \cdot g_i[k-j] = \begin{cases} 1 & k = 0 \\ 0 & k > 0 \text{ and } k < m_i \end{cases} \tag{9}$$

$$\text{Let } G = \begin{bmatrix} g_i[0] & 0 & 0 & 0 & \ldots & 0 \\ g_i[1] & g_i[0] & 0 & 0 & \ldots & 0 \\ g_i[2] & g_i[1] & g_i[0] & 0 & \ldots & 0 \\ g_i[3] & g_i[2] & g_i[1] & g_i[0] & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ g_i[m-1] & g_i[m-2] & g_i[m-3] & g_i[m-4] & \ldots & g_i[0] \end{bmatrix} \tag{10}$$

$$G \begin{bmatrix} p_i[0] \\ p_i[1] \\ p_i[2] \\ p_i[3] \\ \vdots \\ p_i[m-1] \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{11}$$

Let the matrix G_A be such that each element of G_A is the cofactor of the corresponding element from G divided by $\det(G)$. This just means $G^{-1} = G_A^\top$. We also have $det(G) = (g_i[0])^{m_i}$. Rewrite (11) as:

$$\begin{bmatrix} p_i[0] & p_i[1] & \cdots & p_i[m-1] \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} G_A \tag{12}$$

So the coefficients of $p_i(x)$ (what we require) is just the first row of G_A (or in other words the cofactors of the first row of G etc)

There is a nicer representation for $p_i(x)$ however. The determinant of a matrix is the sum of element-cofactor products of (say) the first row. If we replace the first row of G with powers of x i.e.

$(1, x, x^2, \ldots, x^{m_i-1})$ and take the determinant of that matrix, we get $p_i(x)$ since the coefficient of a power of x is the cofactor of that same element in G's first row. We will of-course have to divide by $(g_i[0]^{m_i})$ after taking the determinant. Before all this however, we still need to be able to write $g_i[k]$ in terms of f so we can form the matrix G. We shall do that now

For a general polynomial $P(x)$, let's denote the coefficient of x^k in $P(x)$ as $P[k]$. Also, we denote by $f^k(x)$ the k-th derivative of $f(x)$ where $f^0(x) = f(x)$. From the definition of g_i it will follow that

$$g_i[k] = \frac{f^{k+m_i}(\alpha_i)}{(k+m_i)!} \tag{13}$$

To see why, note that coefficient of x^k in $g_i(x)$ is the same as the coefficient of x^{k+m_i} in $f(x+\alpha_i)$. And that in general $P[k] = P^k(0)/k!$. To introduce some notation, let $F(\alpha, k) := \frac{f^k(\alpha)}{k!}$. So $g_i[k] = F(\alpha_i, k+m_i)$

Thus we have:

$$p_i(x-\alpha_i) = \frac{1}{(F(\alpha_i, m_i))^{m_i}} \begin{vmatrix} 1 & (x-\alpha_i) & (x-\alpha_i)^2 & \ldots & (x-\alpha_i)^{m_i-1} \\ F(\alpha_i, m_i+1) & F(\alpha_i, m_i) & 0 & \ldots & 0 \\ F(\alpha_i, m_i+2) & F(\alpha_i, m_i+1) & F(\alpha_i, m_i) & \ldots & 0 \\ F(\alpha_i, m_i+3) & F(\alpha_i, m_i+2) & F(\alpha_i, m_i+1) & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ F(\alpha_i, 2m_i-1) & F(\alpha_i, 2m_i-2) & F(\alpha_i, 2m_i-3) & \ldots & F(\alpha_i, m_i) \end{vmatrix} \tag{14}$$

For the sake of closure, we can now say that:

$$\frac{1}{f(x)} = \sum_{i=1}^{n} \frac{\det(\Delta[i])}{(F(\alpha_i, m_i)(x-\alpha_i))^{m_i}} \tag{15}$$

where $\Delta[i]$ is a $m_i \times m_i$ matrix defined by:

$$\Delta[i]_{jk} = \begin{cases} (x-\alpha_i)^{k-1} & j = 1 \\ F(\alpha_i, m_i+k-j) & j > 1 \end{cases}$$

The zeros are automatically taken care of in this definition too. Also, when $m_i = 1$, $\Delta[i] = [1]$ and $F(\alpha_i, 1) = f'(\alpha_i)$ so this is consistent with the distinct roots case we started with.

Polynomial Interpolation and the Vandermonde matrix

Suppose I know that a function $H(x)$ takes values β_i at $x = \alpha_i$ for $i \in \{1, 2, \ldots, n\}$ where the α_i are distinct. If I was to give an estimated H from this information assuming that, say, $H(x)$ is a polynomial of degree $n-1$; then $H(x)$ can be 'uniquely' expressed as:

$$H(x) = \sum_{i=1}^{n} \frac{f(x)}{f'(\alpha_i)(x-\alpha_i)} \beta_i \tag{16}$$

Where $f(x) := \prod_{i=1}^{n}(x-\alpha_i)$ for ease of notation (and for relating to the previous formulaes). As before, $f(x)/(x-\alpha_i)$ is just a shorthand for $\prod_{k=1, k\neq i}^{n}(x-\alpha_k)$. The justification for this is also the same as before, $H(x)-$RHS is 0 at n distinct values (the α_i) but is a $n-1$ degree polynomial (at most). So it is zero. These are actually called the Lagrange Interpolating Polynomials.

As a sidenote, we can again extend the same problem for the case of repeated roots. It would be a very convoluted, made-up problem statement but then what isn't? Suppose I know that a function $H(x)$ takes values β_i at $x = \alpha_i$ with a multiplicity m_i for $i \in \{1, 2, \ldots, n\}$ where the α_i are distinct. This means that as a function, H has the property that

$$\lim_{x \to \alpha_i} \frac{H(x) - \beta_i}{(x - \alpha_i)^k} = 0 \qquad \forall k \text{ such that } 0 \leq k < m_i, \forall i \tag{17}$$

If I was to give an estimated H from this information assuming that, say, $H(x)$ is a polynomial of appropriate degree; then $H(x)$ can be expressed as:

$$H(x) = \sum_{i=1}^{n} \frac{p_i(x - \alpha_i) f(x)}{(x - \alpha_i)^{m_i}} \beta_i \tag{18}$$

Where $f(x) := \prod_{i=1}^{n} (x - \alpha_i)^{m_i}$ and $f(x)/(x - \alpha_i)^{m_i}$ would be a shorthand as before. $p_i(x)$ is a polynomial of degree $m_i - 1$ as described in the previous section. The justification for this is also the same as before, $H(x)-$RHS is divisible by $(x - \alpha_i)^{m_i}$ for all i (write in terms of $p(x)g(x) - 1$ to see how). But it's of degree 1 less than $\sum m_i$, so it is zero.

Now a polynomial of degree $n - 1$ is described completely by its n coefficients. But by (16) we can see that its also completely described by the values it returns at some n inputs α_i. Let the coefficient of x^k in $H(x)$ from (16) be c_k where $k \in \{0, 1, \ldots, n-1\}$. The following relation exists between α_i, β_i, c_k:

$$\text{let } V = \begin{bmatrix} 1 & \alpha_1 & \alpha_1^2 & \cdots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \cdots & \alpha_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \cdots & \alpha_n^{n-1} \end{bmatrix} \tag{19}$$

(V stands for Vandermonde)

$$V \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} \tag{20}$$

$$\begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix} = V^{-1} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} \tag{21}$$

From (16) and (21) we conclude that the i-th column of V^{-1} is the set of coefficients of:

$$\frac{f(x)}{f'(\alpha_i)(x - \alpha_i)} = \prod_{k=1, k \neq i}^{n} \frac{x - \alpha_k}{\alpha_i - \alpha_k} \tag{22}$$

Let C be the cofactor matrix of V (each element is the cofactor of the corresponding element in V, and $V^{-1} = C^\top / \det(V)$). Then for example: $C_{i,n}/\det(V) = \prod_{k=1, k \neq i}^{n} \frac{1}{\alpha_i - \alpha_k}$ which is the coefficient of x^{n-1} (calculated in two different ways). Similarly, the coefficient of x^{k-1} for some $k \in \{1, 2, \ldots, n\}$ is (calculated in two different ways):

$$\frac{C_{i,k}}{\det(V)} = \frac{C_{i,n}}{\det(V)} \left(\text{coeff. of } x^{k-1} \text{ in } \prod_{j=1, j \neq i}^{n} (x - \alpha_j) \right) \tag{23}$$

$$C_{i,k} = C_{i,n} \left((-1)^{n-k} \sum_{\substack{S \subseteq U \\ |S| = n-k}} \prod_{j \in S} \alpha_j \right) \qquad U = \{1, 2, \ldots, n\} - \{i\} \text{ and } k \neq n \tag{24}$$

(24) is just an identity involving matrix determinants and is independent of β_i, c_k or H. Also all terms in (24) exclude the variable α_i and it is absent from the equation but it makes it a little cumbersome to think about. $C_{i,k}$ and $C_{i,n}$ are determinants of matrices of size $n-1 \times n-1$. But the same identity will be true for an equivalent $n \times n$ matrix. In short, as far as the underlying identity involving matrix determinants is concerned, it is equivalently: (for new variables x_i with $i \in \{1, 2, \ldots, n\}$)

$$\text{let } V = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{k-1} & x_1^k & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{k-1} & x_2^k & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{k-1} & x_n^k & \cdots & x_n^{n-1} \end{bmatrix}$$

$$\text{let } V_k = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{k-2} & x_1^k & \cdots & x_1^{n} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{k-2} & x_2^k & \cdots & x_2^{n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{k-2} & x_n^k & \cdots & x_n^{n} \end{bmatrix} \qquad 1 \le k \le n$$

$$\det(V_k) = \det(V) \left(\sum_{\substack{S \subseteq U \\ |S| = n-k+1}} \prod_{j \in S} x_j \right) \qquad U = \{1, 2, \ldots, n\}$$

Here V is a standard Vandermonde matrix and V_k is formed by removing the k-th column from V and adding a new column at the right end for the next power (n).

'Partial fractions' for non-polynomials

Though this section is quite hand-wavy, most results do happen to be correct and well-known.

To decompose something into partial fractions as in (2) we just need to know the roots of $f(x)$ and the derivative of f at those roots. Of course, $f(x)$ needs to be a polynomial too but what if I decide to humour the formulae? What if, say, $f(x) = \sin x$?

Think of $\sin x$ as a polynomial with roots $n\pi$ for all integers n. We may try to compare $\sin x$ to a more polynomial form of $x(x^2 - \pi^2)(x^2 - 4\pi^2) \cdots$. This doesn't look right because we probably need to multiply by some appropriate constant so that, say $\lim_{x \to 0}(\sin x / x) = 0$. So maybe we compare $\sin x$ to a 'polynomial' that looks like: $x \prod_{k=1}^{\infty} \left(1 - \left(\frac{x}{k\pi}\right)^2\right)$. This does come close to $\sin x$ which you can see by trying to graph a finite product. This is actually known as Weierstrass factorization. So what if we pretend this is $\sin x$ to see how $1/\sin x$ will break into partial fractions according to (2)? We get

$$\frac{1}{\sin x} = \sum_{k=-\infty}^{\infty} \frac{(-1)^k}{x - k\pi} \tag{25}$$

This funny looking equation is actually true, though more commonly written in the following equivalent format:

$$\frac{x \csc x - 1}{2x^2} = \sum_{k=1}^{\infty} \frac{(-1)^k}{x^2 - k^2\pi^2} \tag{26}$$

But there is an important issue with our line of reasoning so far. Other than the lack of rigour, of course. We seem to imply that any function $f(x)$ who equals zero only at $x = n\pi$ and has the derivative as $f'(n\pi) = (-1)^n$ will be able to take the place of $\sin x$ in (25). But clearly there are functions other than $\sin x$ that are zero at $n\pi$ with slope $(-1)^n$. You can construct any such example but let's take a simple one: $\sin x - a \sin^2 x$ for some $a \in (0, 1)$

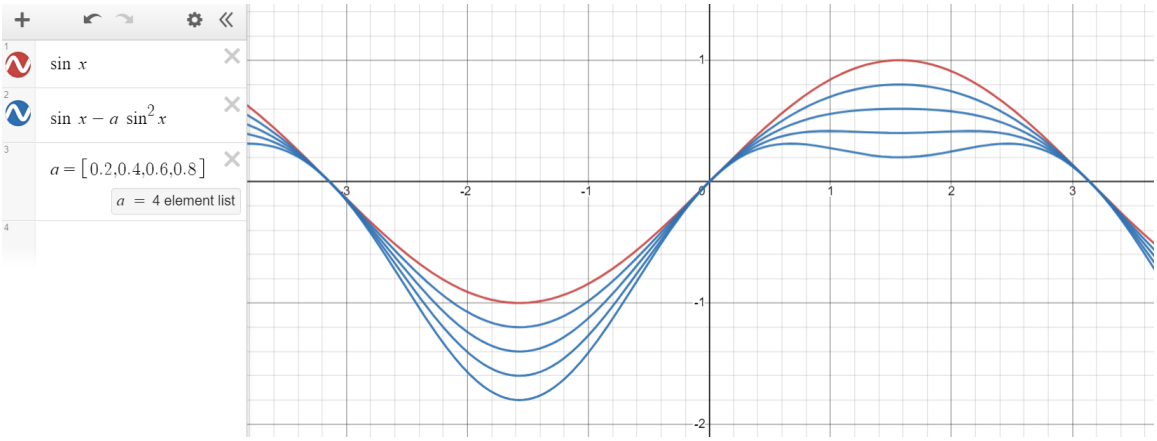

Figure 1: $(\sin x)$ vs $(\sin x - a\sin^2 x)$

Let $f_a(x) = \sin x - a\sin^2 x$. If we are going to pretend $f_a(x)$ and $f(x)$ are polynomials, then we need to be including their non-real roots too. If we define $\sin x$ for $x \in \mathbb{C}$ as $\frac{e^{ix}-e^{-ix}}{2i}$ then $f(x) = \sin x$ still has only real roots but $f_a(x)$ has some additional non real roots (from solving $\sin x = 1/a$) that we need to have corresponding terms for in the partial fraction decomposition. In other words, when we say that there are no roots other than $n\pi$, that includes non-real roots.

If (25) is correct, it seems to also imply that the exact shape of $\sin x$ is just the 'right' shape for starting at $x = 0$ with a slope of 1 and finishing at $x = \pi$ with a slope of -1, for example. If we try to do this by drawing any other shape there will apparently be some underlying non-real roots associated with it.

To test our claim about taking into account the non-real roots of $f_a(x)$ too in the partial fraction decomposition, we calculate that these are of the form

$$x_k = \alpha_k \mp i\beta \qquad\qquad k \in \mathbb{Z} \tag{27}$$

where

$$\alpha_k = \frac{\pi}{2} + 2k\pi \tag{28}$$

$$\beta = \ln\left(\frac{1}{a} + \sqrt{\frac{1}{a^2} - 1}\right) \tag{29}$$

and

$$f_a'(x_k) = \mp i\sqrt{\frac{1}{a^2} - 1} \tag{30}$$

So the additional terms due to non real roots will be:

$$\frac{1}{\left(i\sqrt{\frac{1}{a^2}-1}\right)(x - (\alpha_k + i\beta))} + \frac{1}{\left(-i\sqrt{\frac{1}{a^2}-1}\right)(x - (\alpha_k - i\beta))} = \frac{\frac{2\beta a}{\sqrt{1-a^2}}}{x^2 + 2\alpha_k x + \alpha_k^2 + \beta^2} \tag{31}$$

And therefore we want to claim that:

$$\frac{1}{\sin x - a\sin^2 x} = \sum_{k=-\infty}^{\infty} \frac{(-1)^k}{x - k\pi} + \sum_{k=-\infty}^{\infty} \frac{\frac{2\beta a}{\sqrt{1-a^2}}}{x^2 + 2\alpha_k x + \alpha_k^2 + \beta^2} \tag{32}$$

I can't 'prove' this equality but approximating finite sums on any graphing calculator can show us that they are close enough and so (probably) equal.

So, if it works for $\sin x$ then it probably works for some other simple functions too. If $f(x) = \tan x$ then f is zero at only $n\pi$ (including possible non-real roots) for $n \in \mathbb{Z}$ and $f'(n\pi) = 1$. So

$$\frac{1}{\tan x} = \sum_{k=-\infty}^{\infty} \frac{1}{x - k\pi} \tag{33}$$

which is also true and so is the case of $f(x) = \cos x$ and $f(x) = \cot x$ etc. Apparantly the discontinuity at $x = k\pi + \pi/2$ when treating $f(x) = \tan x$ as a polynomial does not break (33). The wikipedia example of $\csc^2 x$ suggests that the equations for the repeated roots case also hold for 'non-polynomials'. I claim, without proof, that (2) and (15) are awesome.